cloudspaces

D2.2: Architecture

Pedro García López

Universitat Rovira i Virgili

SEVENTH FRAMEWORK PROGRAMME

UNIVERSITAT ROVIRA I VIRGILI

Contents

- StackSync Framework
- Adaptive Personal Storage
- Privacy-aware data sharing
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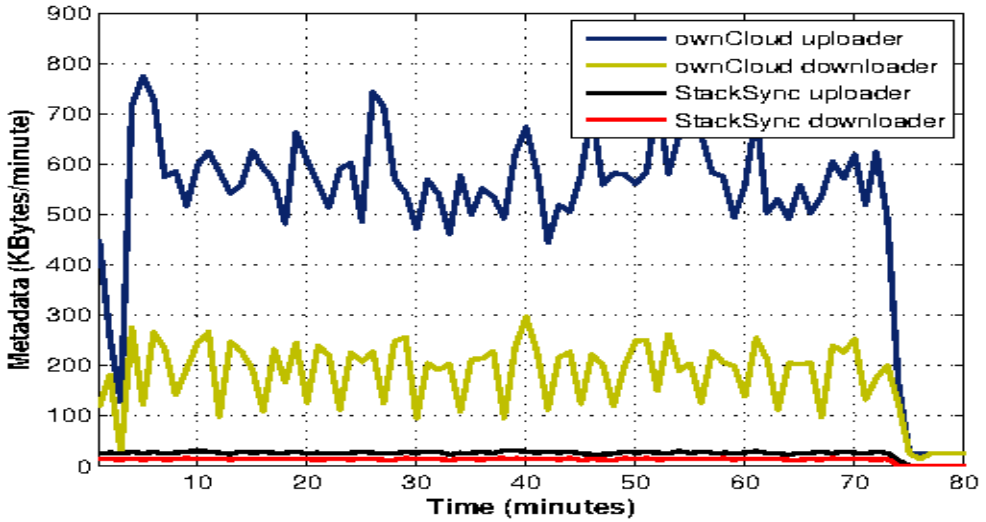


Why StackSync

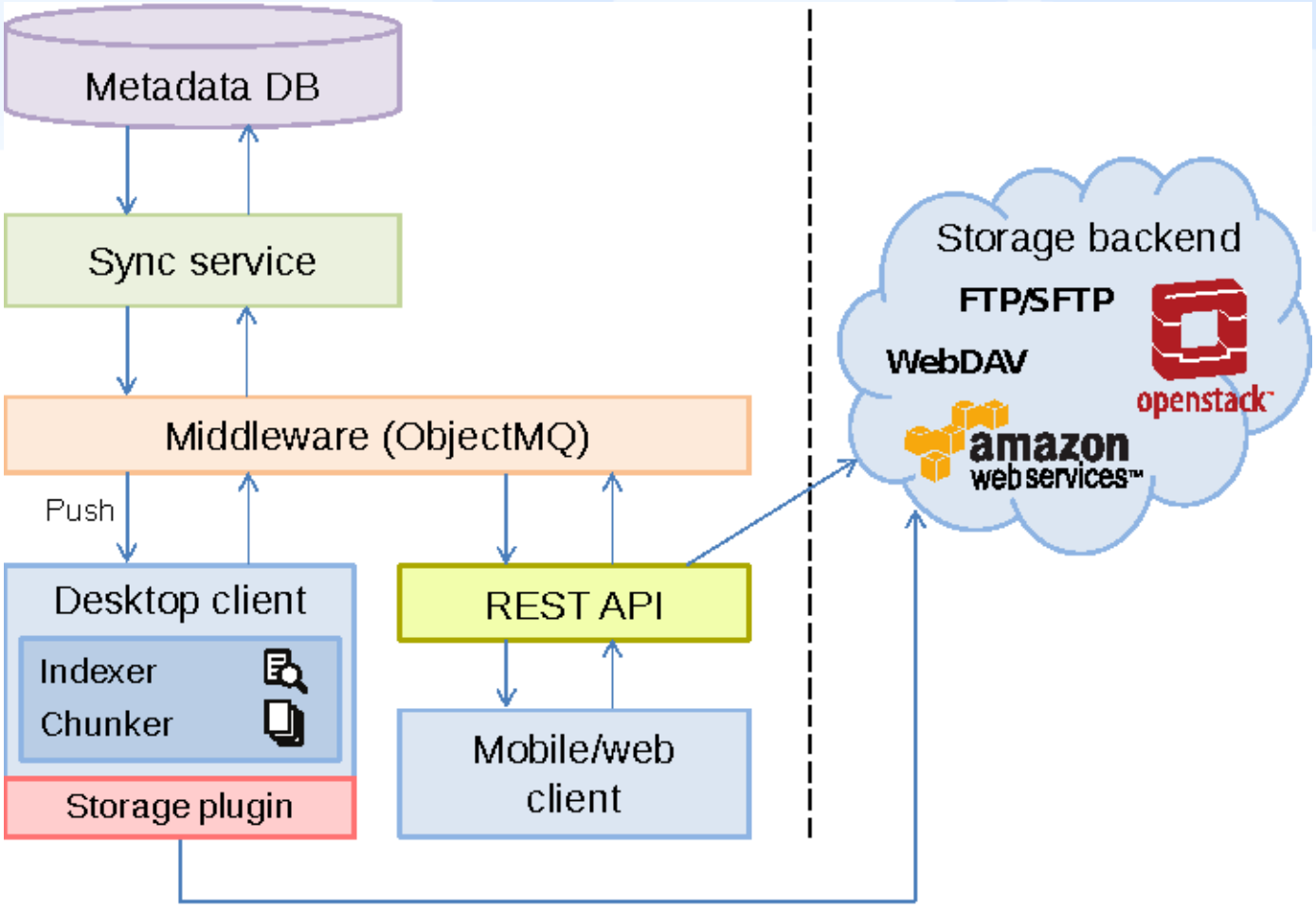
Clear need for an open source Personal Cloud:

Existing systems like ownCloud lack important aspects:

- Modularity and extensibility
- Scalability and reliability
- Privacy and Interoperability



StackSync Architecture



StackSync Architecture

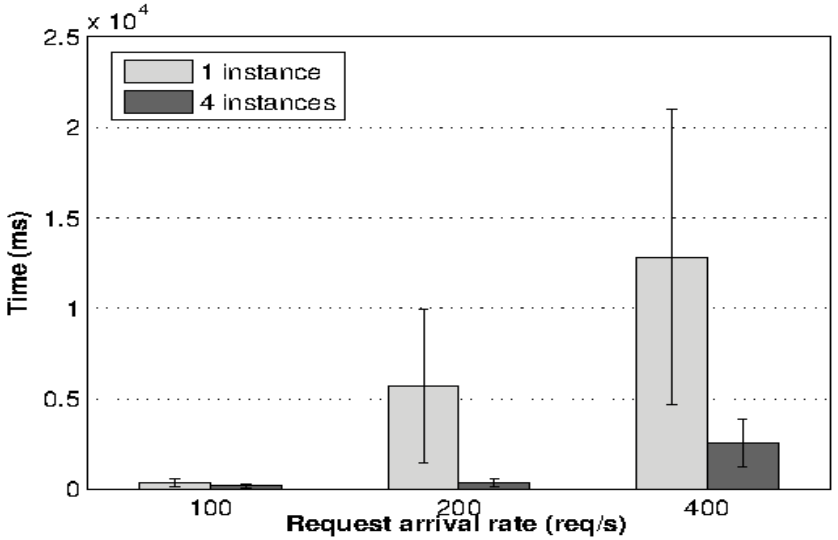
Push Model for change notifications

Use of scalable cloud storage (OpenStack Swift)

Modular plugin-based architecture

Separates metadata from storage backend

Designed to be elastic (ObjectMQ)

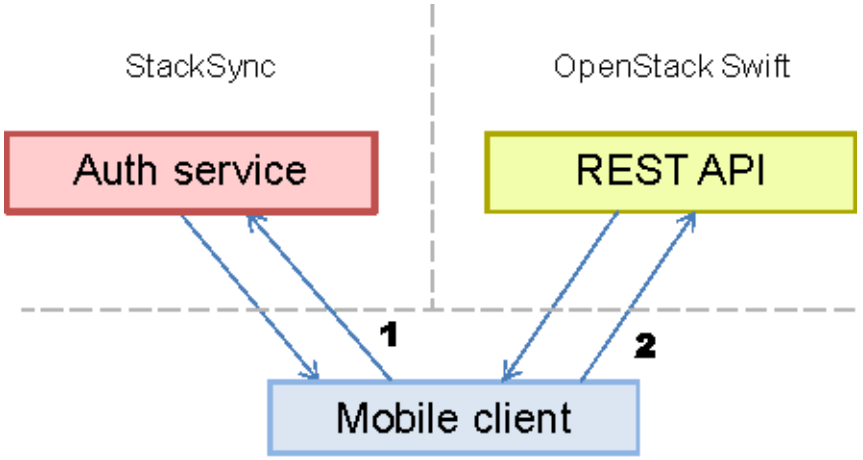


StackSync Interoperability

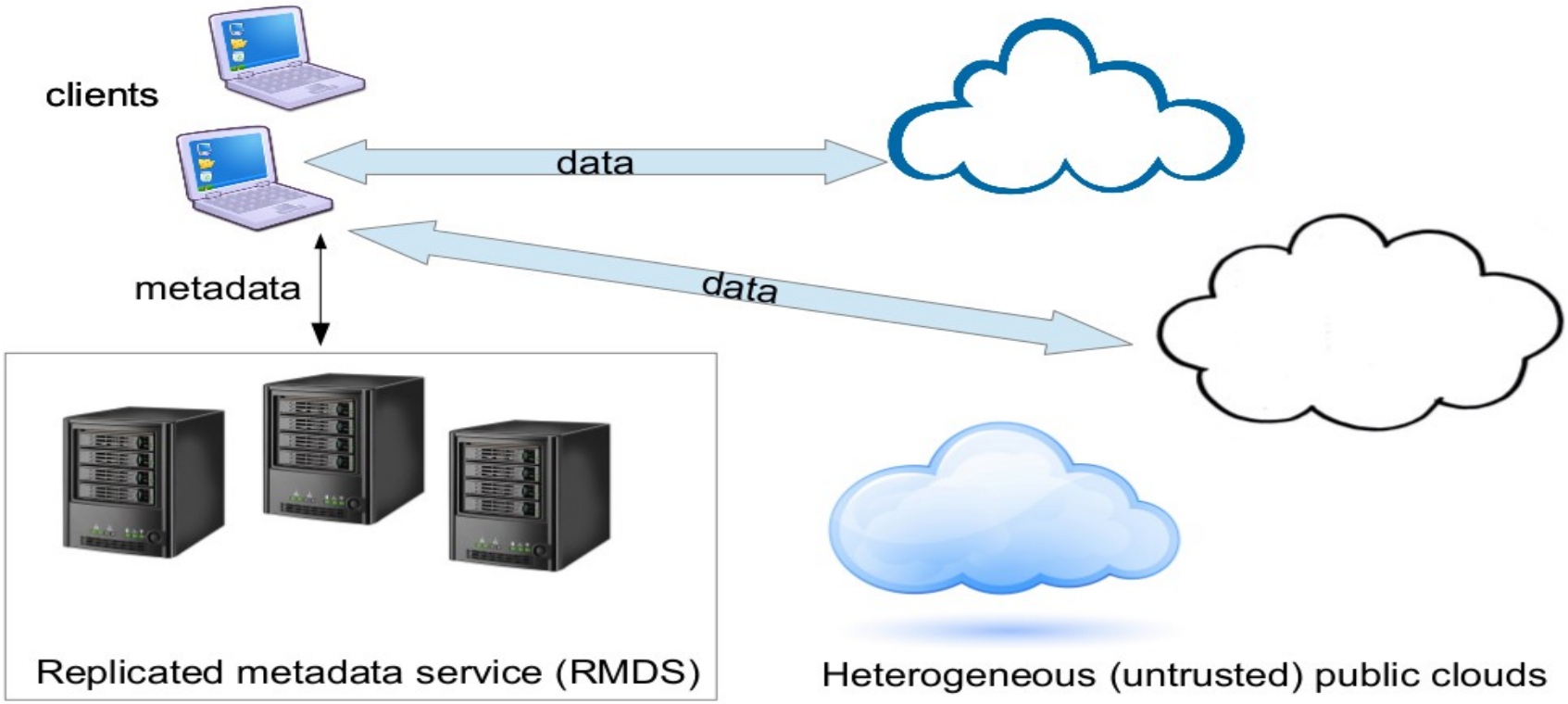
REST APIs for Storage

First applications using APIs: eyeOs, Mobile, ownCloud Web

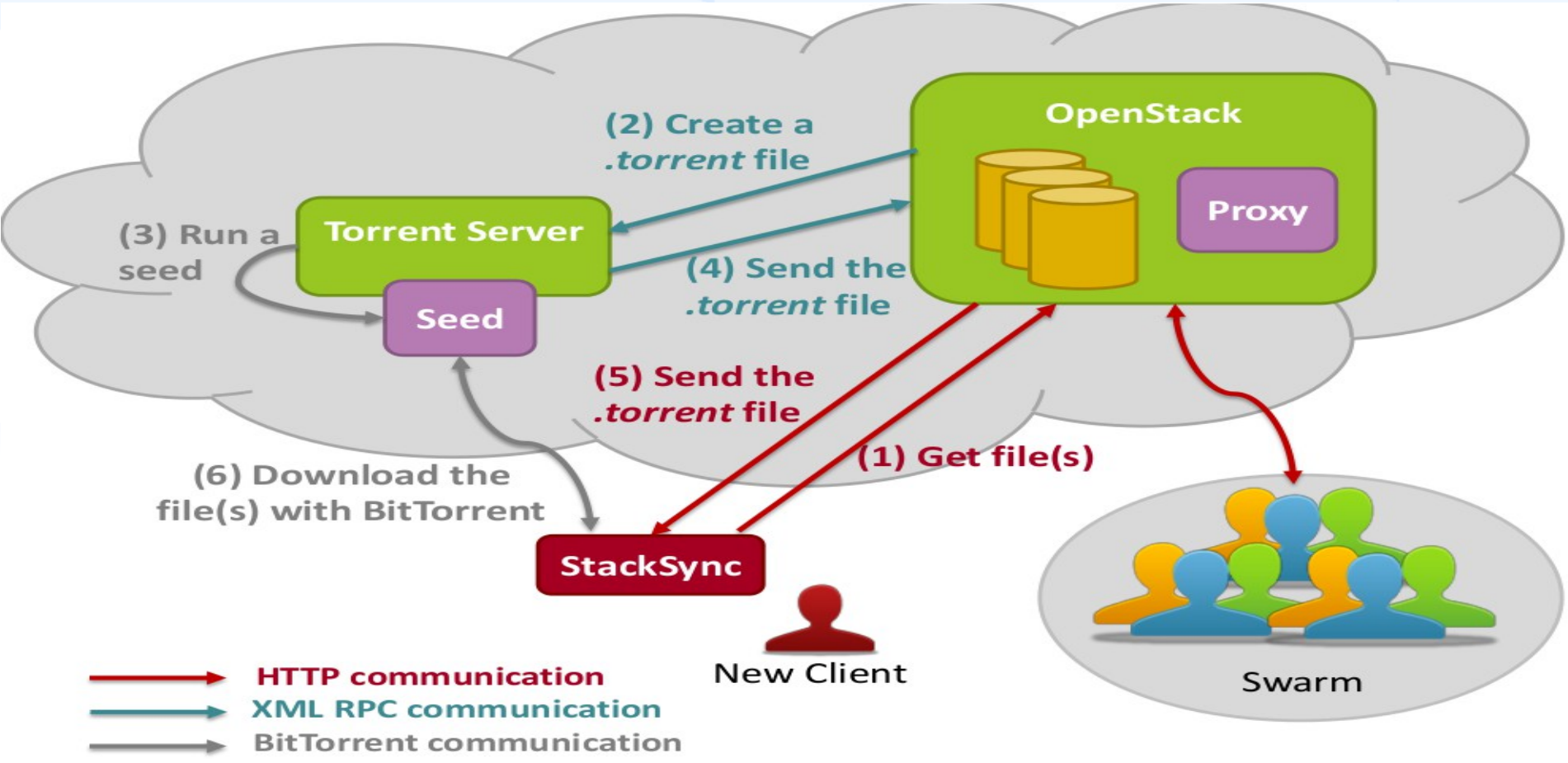
Authentication using OAUTH modules in OpenStack



Hybrid Cloud Storage

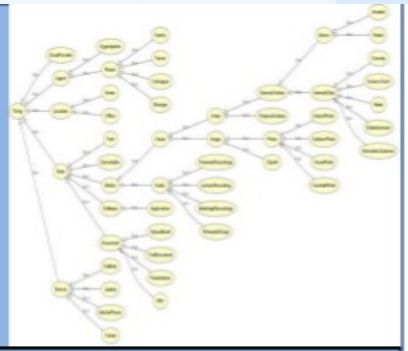


Adaptive Content Distribution



Privacy-aware data sharing

Vocabulary of shared data



Semantic Policy Crowd-Sourcer

Aggregation of various users policies for shared data

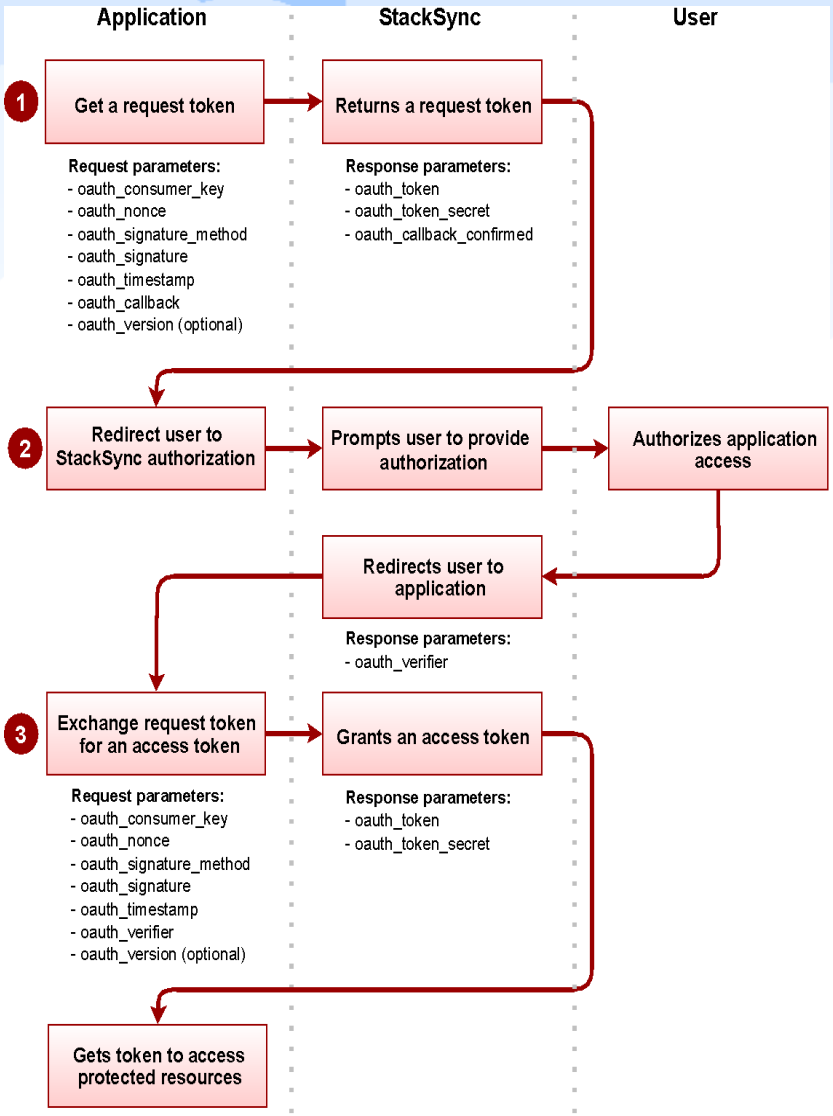
Automated Risk Prediction Based on Psychometric Models

Risk Evaluation Component

Other facets such as Trust

Service Platform

- Authentication, Store, Share, Proximity, Portability, Persistence
- Oauth-based protocol for authentication and sharing

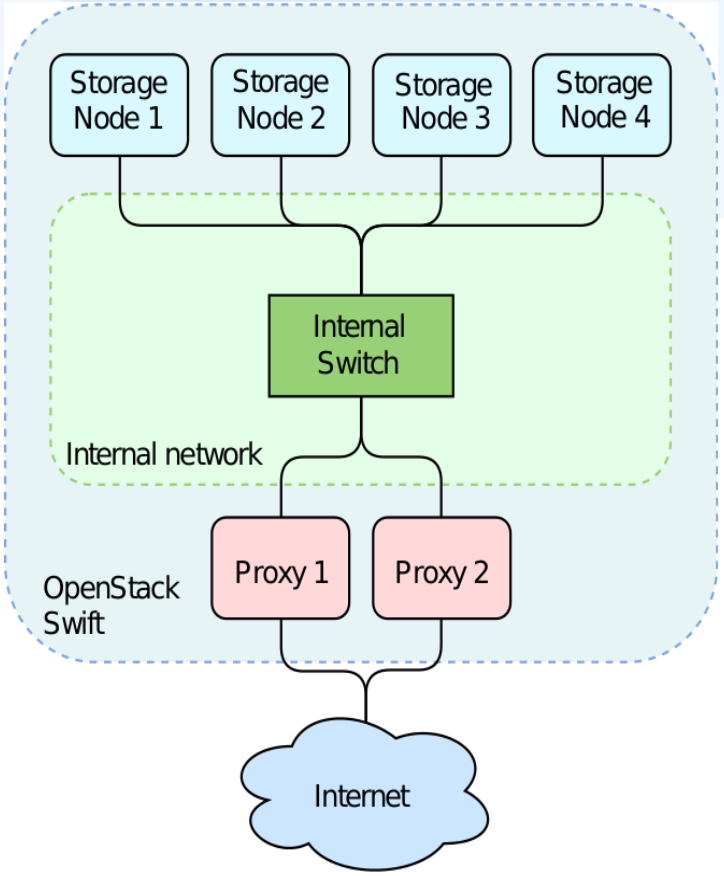


Expected Evaluation

- Ubuntu One dataset
- EyeOS dataset
- StackSync dataset
- Enron dataset
- YCSB (Yahoo Cloud Serving Benchmark)
- IMC 2013, Benchmarking Personal Cloud Storage

Testbeds

- URV testbed
  - OpenStack Swift/StackSync
  - Rack at URV
- TISSAT Testbed
  - OpenStack Swift/StackSync
  - Tier 4 datacenter Walhalla
- Canonical Testbed
  - Alternative U1 site
  - Interoperability proxy



Measuring Personal Cloud Storage

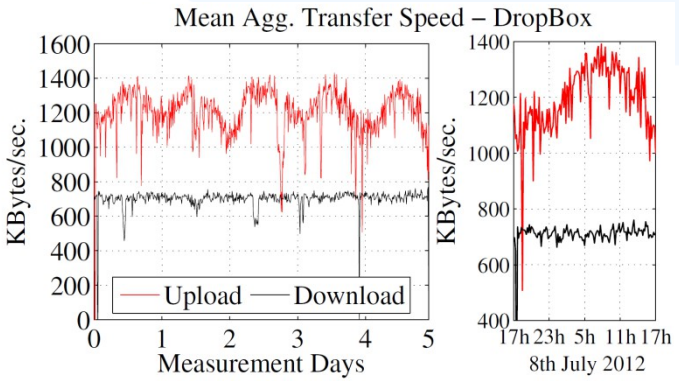
Public measurement from Planetlab network of Dropbox, Box, and SugarSync APIs (70TB of data). Results:

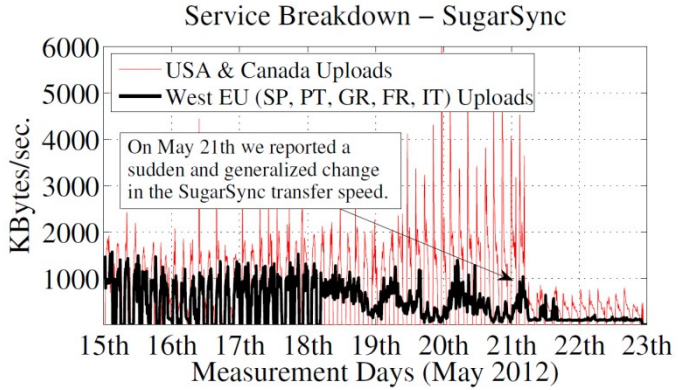
- ***API Services are heterogeneous**: There exist significant differences among providers in the QoS of their REST APIs.*

- ***API Services are (really) variable**: transfer speeds vary depending on the client's Geographic location, the file size, the traffic type (in/out) or the hour of the day (daily patterns).*

- ***QoS changes and data lock-in**: We proved that SugarSync changed its freemium QoS unexpectedly (importance of storage diversity).*

- ***These APIs are in general reliable**.*





Demonstrations

- StackSync
  - Clients (URV)
  - Web Management (TST)
- Horizontal Interoperability (CNC)
- Vertical Interoperability (EOS)
- Privacy demo (EPFL)
- Adaptive Storage
  - Hybrid Storage (EUR)
  - Adaptive Content Distribution (URV)







